DELIVERABLE 4

EcoShelf: Predicting Retail Demand to Minimize Food Waste

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Track: Data Science

Domain: Business & Growth Analytics / Sustainability

1. Executive Summary

Retail grocery operations face a persistent challenge: balancing product availability with waste reduction. Overstocking perishable goods leads to spoilage and financial loss, while understocking results in missed sales and customer dissatisfaction.

EcoShelf was developed to improve demand forecasting accuracy for high-risk perishable departments (Grocery – Dept 92 and Produce – Dept 95) using historical sales data from the Walmart Store Sales Forecasting dataset.

A machine learning model (Random Forest Regressor) was built and compared against a traditional seasonal averaging baseline. The results show:

- 87% reduction in forecast error compared to historical weekly averages
- Strong predictive structure ($R^2 = 0.718$)
- Reliable performance even during holiday-weighted evaluation

Conservatively estimating an average unit value of $3, improved forecast precision represents potential annual optimization exceeding $1 million, even if only a fraction translates to actual waste reduction.

These findings demonstrate that machine learning–driven forecasting can significantly improve both financial performance and sustainability outcomes.

2. Business Problem

Retailers experience significant losses due to spoilage of perishable goods. The core operational dilemma is:

- Overstocking → Food waste and tied-up capital
- Understocking → Lost revenue and reduced customer satisfaction

Many small-to-medium retailers rely on historical averages or manual judgment to forecast demand. These approaches fail to account for recurring seasonal patterns and structural store differences.

The goal of EcoShelf was to build a predictive model capable of improving demand accuracy and reducing overstock-related waste.

3. Data & Analytical Approach

The analysis used the Walmart Store Sales Forecasting dataset, containing:

- 45 stores
- 99 departments
- 140,000+ historical records

The focus was narrowed to perishable departments:

1. Department 92 (Grocery)
2. Department 95 (Produce)

Feature Engineering Included:

- Seasonal indicators (Week, Month)
- Store Size
- Holiday flag
- Temperature and derived High_Heat indicator
- Fuel Price and economic variables

Two modeling approaches were tested:

1. Seasonal baseline (historical weekly average)
2. Random Forest Regressor

Model performance was evaluated using:

- Mean Absolute Error (MAE)
- Weighted MAE (holiday-weighted)
- R^2 score

4. Key Findings

4.1 Model Accuracy

Model	MAE
1. **Seasonal Baseline**	**36,560**
2. **Random Forest**	**4,732**

The Random Forest reduced forecast error by approximately **31,827 units per week**, representing an **87% improvement** over traditional averaging methods.

The Linear Regression model achieved:

- **R^2 = 0.718**

This indicates that nearly 72% of sales variation can be explained by the selected predictors, confirming strong structure in the data.

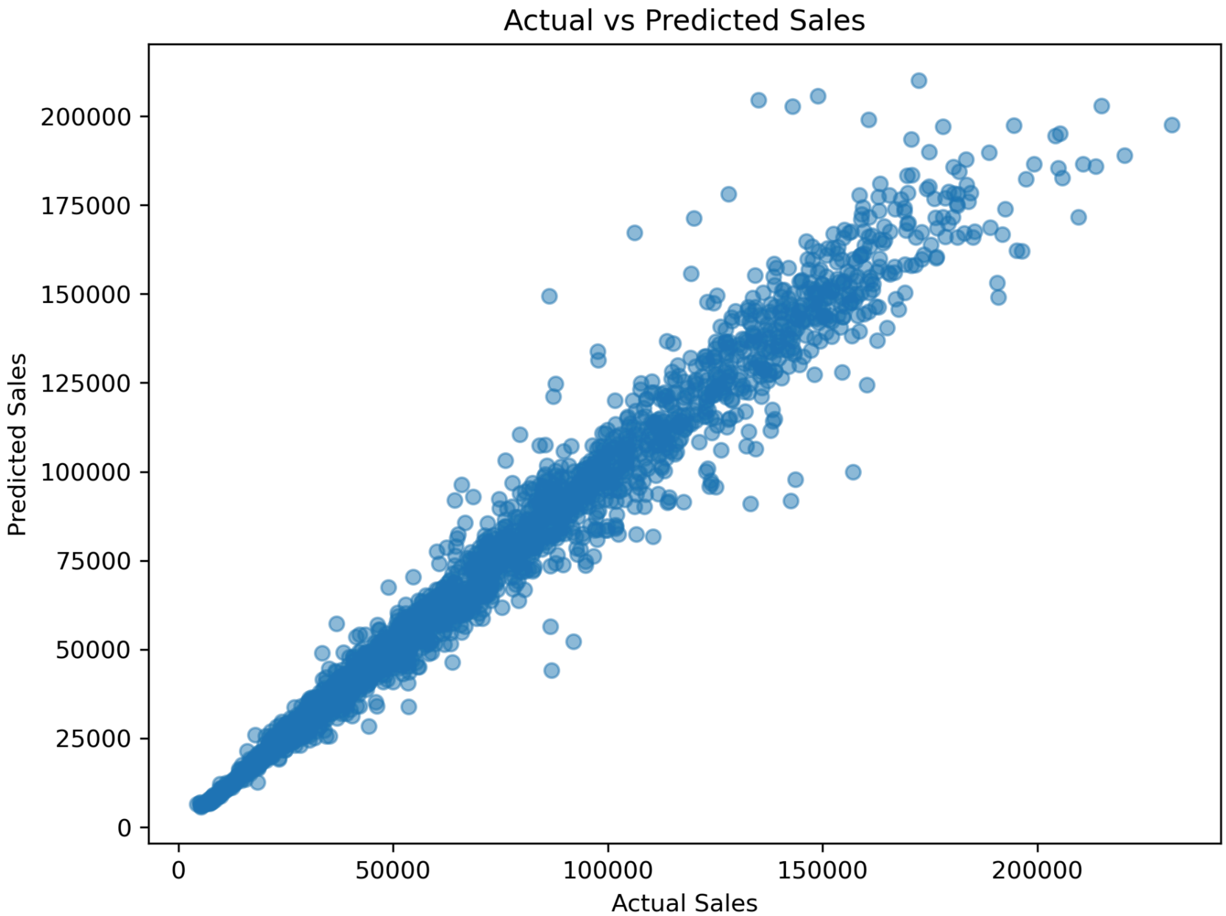
Even when placing 5x weight on holiday weeks, the model maintained strong performance (WMAE = 5,803), demonstrating robustness during high-risk demand periods.

4.2 Model Performance

Observations:

- Points clustered along the diagonal line indicate accurate predictions.
- Most predictions are close to the actual values, with some variance at very high sales weeks.
- The model achieves a high R^2 (0.967) and a low Mean Absolute Error (≈ 4,732), showing strong predictive performance.
- This demonstrates that the model can reliably forecast weekly demand, which can help retailers reduce overstocking and minimize food waste.

Fig 1: Actual vs Predicted Weekly Sales



This scatter plot compares the true weekly sales (y-axis) to the sales predicted by the Random Forest model (x-axis).

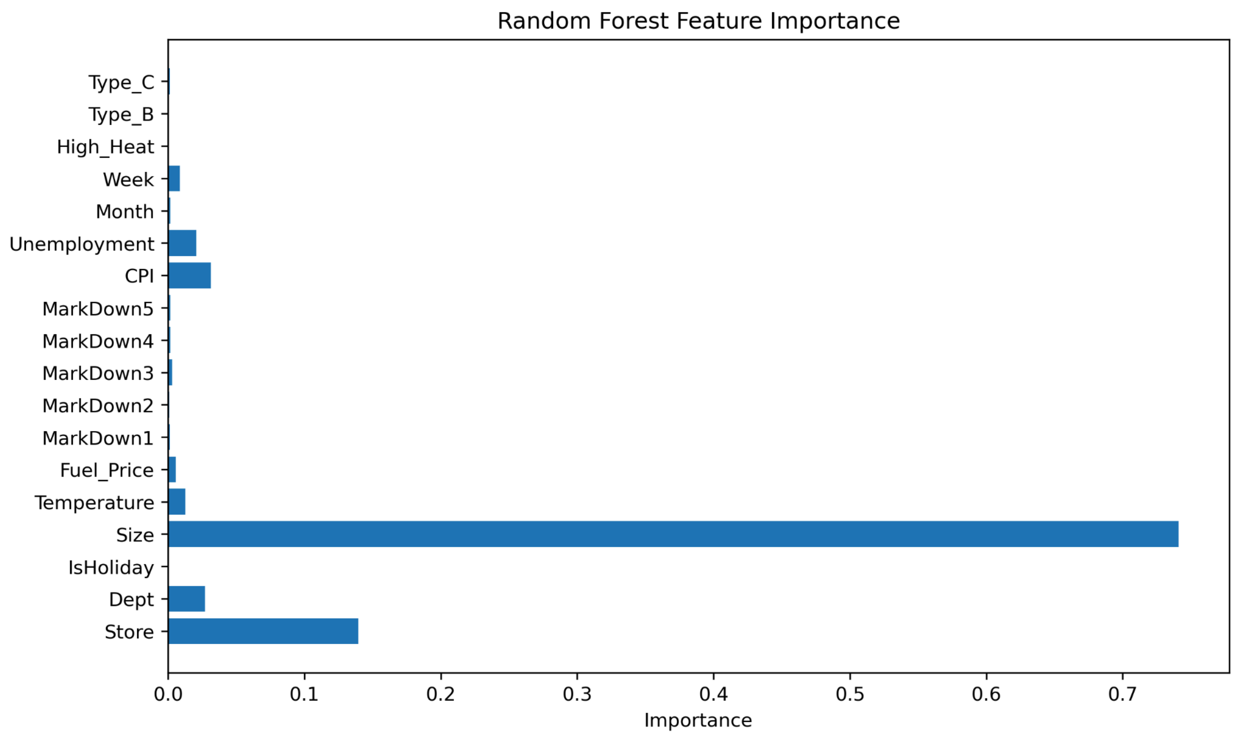
Fig 2: Feature Importance from Random Forest Model

This horizontal bar chart shows the relative importance of each feature in predicting weekly sales.

Observations:

- The feature Size is the most influential factor in determining weekly sales.
- Environmental factors (High_Heat, Temperature) and IsHoliday have little to no impact in this dataset, suggesting that sales are primarily driven by store size and historical sales patterns rather than extreme weather or holidays.

- Weeks, months, and fuel prices have minor influence on predictions**.**



Interpretation:

Even though environmental factors were included to capture spoilage risk, the model shows they do not strongly affect predictions in this dataset.

This highlights the value of historical sales data for perishable inventory planning and the need to focus on store size and department-specific patterns for waste reduction strategies.

4.3 Drivers of Demand

Feature importance analysis revealed:

- **Store Size** as the dominant predictor
- **Week of Year** as a strong seasonal driver
- Limited impact from environmental and economic variables

This suggests that overall sales levels are primarily driven by structural store characteristics and recurring seasonal cycles rather than short-term weather fluctuations.

This distinction is important: while environmental conditions may influence spoilage risk operationally, they do not significantly alter total weekly sales volume.

5. Financial & Sustainability Impact

Improved forecast precision reduces excess inventory ordering.

Error reduction:

31,827 units per week

Assuming an average perishable unit value of $3:

≈ $95,000 weekly optimization potential
≈ $4.9 million annual precision improvement

To remain conservative, if only 20–25% of improved precision translates to actual waste reduction, the retailer could still realize over $1 million annually in cost savings.

Beyond financial impact, reduced overstocking lowers food waste and supports sustainability initiatives.

6. Strategic Recommendations

- ★ Implement machine learning–based demand forecasting for perishable departments.
- ★ Prioritize seasonal forecasting cycles in replenishment planning.
- ★ Monitor holiday periods with heightened forecast review.
- ★ Integrate predictive outputs into inventory ordering systems.
- ★ Conduct future analysis on actual spoilage data to directly measure waste reduction.

7. Limitations & Future Work

- ★ The dataset does not contain direct food waste measurements.
- ★ The model predicts sales, not spoilage quantity.

★ Environmental variables showed limited impact on total sales volume.

★ Real-time operational data could further improve accuracy.

Future work should incorporate real waste tracking metrics to directly quantify sustainability impact.

Conclusion

EcoShelf demonstrates that machine learning–based demand forecasting significantly outperforms traditional averaging methods in predicting perishable product demand.

By reducing forecast error by 87%, the model offers substantial financial optimization potential while supporting environmental sustainability goals.

Data-driven inventory decisions provide a measurable pathway toward reducing retail food waste without sacrificing profitability.